H524 Introduction to Biostatistics

Week 3: Theoretical Probability Distributions and Sampling Distributions

Dr. John Molitor

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Outline

Quick Review: Probability Concepts

Last week we covered:

- Basic probability rules and calculations
- Conditional probability: $\Pr(A \mid B)$
- Independence: $\Pr(A \cap B) = \Pr(A) \times \Pr(B)$
- Diagnostic testing: sensitivity, specificity, PPV, NPV
- Relative risk and odds ratios

This Week: Theoretical probability distributions and their role in statistical inference

Reading: Pagano & Gauvreau Chapter 7 (7.1, 7.2, 7.4) and Chapter 8

Why Probability Distributions?

From Week 2: We calculated probabilities for specific events

This Week: Use theoretical models to describe entire families of outcomes

Advantages of theoretical distributions:

- Predict probabilities without collecting all possible data
- Foundation for statistical inference (hypothesis testing, confidence intervals)
- Standardized models applicable across many scenarios
- Enable calculation of complex probabilities efficiently

Two main types:

- **Discrete distributions:** Countable outcomes (binomial, Poisson)
- **Continuous distributions:** Uncountable outcomes (normal, t, chi-square)

Binomial Distribution: Conditions

A binomial experiment must satisfy:

1. **Fixed number of trials:** n is predetermined
2. **Two possible outcomes:** Success or failure
3. **Constant probability:** p remains the same for all trials
4. **Independent trials:** Outcome of one trial doesn't affect others
5. **Random variable:** $X =$ number of successes in n trials

Notation: $X \sim \text{Binomial}(n, p)$ or $X \sim B(n, p)$

Medical Example:

- $n = 20$ patients receive a vaccine
- $p = 0.90$ probability vaccine protects (efficacy)
- $X =$ number of patients protected

Binomial Probability Mass Function

Probability of exactly k successes:

$$\Pr(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the binomial coefficient

Parameters and Properties:

- Expected value (mean): $\mu = E(X) = np$
- Variance: $\sigma^2 = \text{Var}(X) = np(1-p)$
- Standard deviation: $\sigma = \sqrt{np(1-p)}$
- Range: $X \in \{0, 1, 2, \ldots, n\}$

Binomial Example: Clinical Trial

Scenario: A new treatment has 75% success rate. We treat 10 patients.

Question 1: What's the probability exactly 8 patients respond?

$$\Pr(X = 8) = \binom{10}{8}(0.75)^8(0.25)^2$$
$$= 45 \times 0.1001 \times 0.0625 = 0.282$$

Question 2: What's the probability at least 8 respond?

$$\Pr(X \geq 8) = \Pr(X = 8) + \Pr(X = 9) + \Pr(X = 10)$$
$$= 0.282 + 0.188 + 0.056 = 0.526$$

Expected number of successes: $\mu = 10 \times 0.75 = 7.5$ patients

Carcinogen Exposure Example

From Pagano & Gauvreau: 25% of people exposed to a carcinogen become ill. Among 4 people with equal exposure:

Calculate probabilities for 0, 1, 2, 3, 4 people becoming ill:

k	Calculation	Probability	Fraction
0	$\binom{4}{0}(0.25)^0(0.75)^4$	0.316	81/256
1	$\binom{4}{1}(0.25)^1(0.75)^3$	0.422	108/256
2	$\binom{4}{2}(0.25)^2(0.75)^2$	0.211	54/256
3	$\binom{4}{3}(0.25)^3(0.75)^1$	0.047	12/256
4	$\binom{4}{4}(0.25)^4(0.75)^0$	0.004	1/256

Binomial Distribution in R

Four key functions:

- `dbinom(k, n, p)` – probability of exactly k successes
- `pbinom(k, n, p)` – cumulative probability $\Pr(X \leq k)$
- `qbinom(q, n, p)` – quantile function (inverse of pbinom)
- `rbinom(m, n, p)` – generate m random binomial values

Example: Treatment with $n = 10$, $p = 0.75$

- `dbinom(8, 10, 0.75)` returns 0.282
- `pbinom(7, 10, 0.75)` returns $\Pr(X \leq 7) = 0.474$
- `pbinom(7, 10, 0.75, lower.tail=FALSE)` returns $\Pr(X > 7) = 0.526$
- `1 - pbinom(7, 10, 0.75)` also returns 0.526

From Discrete to Continuous

Continuous Random Variables:

- Can take any value in an interval
- Examples: height, weight, blood pressure, cholesterol
- Probability described by a *density curve*
- $\Pr(a \leq X \leq b) =$ area under curve between a and b

Key properties of density curves:

- Curve never goes below the x-axis
- Total area under curve equals 1
- $\Pr(X = c) = 0$ for any single point c
  - Therefore $\Pr(a \leq X \leq b) = \Pr(a < X < b)$

The Normal Distribution

Most important continuous distribution in statistics

Characteristics:

- Bell-shaped, symmetric curve
- Completely determined by mean μ and standard deviation σ
- Notation: $X \sim N(\mu, \sigma^2)$
- Theoretical range: $-\infty$ to $+\infty$

Important: Not all bell-shaped curves are normal!

- t-distribution: similar shape but heavier tails
- Chi-square: right-skewed for small df
- Many biological variables are approximately normal

The Empirical Rule (68-95-99.7)

For any normal distribution:

- Approximately **68%** of observations lie within **1 SD** of the mean

$$\Pr(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68$$

- Approximately **95%** lie within **2 SDs** of the mean

$$\Pr(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$

- Approximately **99.7%** lie within **3 SDs** of the mean

$$\Pr(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997$$

Application: Quick assessment of unusual observations

- Values beyond 2 SDs: unusual (only 5% of data)
- Values beyond 3 SDs: very unusual (only 0.3% of data)

Standard Normal Distribution

Special case: $Z \sim N(0, 1)$ (mean $= 0$, SD $= 1$)

Why is this useful?

- We can standardize ANY normal distribution to $N(0, 1)$
- Only need one table (Table A.3) for all normal probabilities
- Foundation for hypothesis testing and confidence intervals

Standardization (Z-score transformation):

$$Z = \frac{X - \mu}{\sigma}$$

If $X \sim N(\mu, \sigma^2)$, then $Z \sim N(0, 1)$

Interpretation: Z-score tells you how many SDs X is from the mean

- $Z = 1.5$ means X is 1.5 SDs above the mean
- $Z = -2.0$ means X is 2 SDs below the mean

Working with the Standard Normal

Using Table A.3 (or R):

If $Z \sim N(0, 1)$:

- $\Pr(Z \leq 0) = 0.5$ (symmetry)
- $\Pr(Z < 1) = 0.8413$
- $\Pr(Z < -1) = 0.1587$
- $\Pr(Z > 1.96) = 1 - 0.975 = 0.025$
- $\Pr(-1.96 < Z < 1.96) = 0.95$ (important for CIs!)

R Functions:

- `pnorm(z)` returns $\Pr(Z \leq z)$
- `qnorm(p)` returns z such that $\Pr(Z \leq z) = p$
- `dnorm(z)` returns the density at z (for plotting)

Normal Distribution: Exam Scores Example

Biostatistics final exam scores: $\mu = 70$, $\sigma = 10$

So $S \sim N(70, 10^2)$

Question 1: What proportion of students score above 85?

Solution:

1. Standardize: $Z = \frac{85-70}{10} = 1.5$
2. Find probability: $\Pr(S > 85) = \Pr(Z > 1.5)$
3. From table: $\Pr(Z < 1.5) = 0.9332$
4. Therefore: $\Pr(Z > 1.5) = 1 - 0.9332 = 0.0668$

Answer: About 6.7% of students score above 85

R code: `pnorm(85, mean=70, sd=10, lower.tail=FALSE)`

Normal Distribution: Between Two Values

Same exam: What proportion score between 60 and 85?

Solution:

1. Standardize lower bound: $Z_1 = \frac{60-70}{10} = -1.0$
2. Standardize upper bound: $Z_2 = \frac{85-70}{10} = 1.5$
3. $\Pr(60 < S < 85) = \Pr(-1 < Z < 1.5)$
4. $= \Pr(Z < 1.5) - \Pr(Z < -1.0)$
5. $= 0.9332 - 0.1587 = 0.7745$

Answer: About 77.5% score between 60 and 85

R code: `pnorm(85, 70, 10) - pnorm(60, 70, 10)`

Finding Percentiles

Question: What score is at the 90th percentile? (Top 10%)

Solution - Working backwards:

1. We want S_{90} such that $\Pr(S < S_{90}) = 0.90$
2. Standardize: $\Pr\left[Z < \frac{S_{90}-70}{10}\right] = 0.90$
3. From table: $\Pr(Z < 1.28) = 0.90$
4. Therefore: $\frac{S_{90}-70}{10} = 1.28$
5. Solve: $S_{90} = 70 + 1.28(10) = 82.8$

Answer: Score of 82.8 is the 90th percentile

R code: `qnorm(0.90, mean=70, sd=10)` returns 82.8

Medical Example: Cholesterol Levels

Adult cholesterol: $\mu = 200$ mg/dL, $\sigma = 40$ mg/dL

Clinical thresholds:

- < 200 mg/dL: Desirable
- $200 - 239$ mg/dL: Borderline high
- ≥ 240 mg/dL: High (treatment recommended)

Question: What proportion need treatment (cholesterol ≥ 240)?

Solution:

- $Z = \frac{240-200}{40} = 1.0$
- $\Pr(X \geq 240) = \Pr(Z > 1.0) = 1 - 0.8413 = 0.1587$

Answer: About 16% of adults have high cholesterol requiring treatment

Student's t-Distribution

Named after "Student" (William Sealy Gosset)

- Brewer and statistician at Guinness
- Published under pseudonym "Student" (1908)
- Developed for small sample inference

Properties:

- Bell-shaped like normal, but heavier tails
- Determined by degrees of freedom (df = n - 1)
- As df increases, approaches standard normal
- Used when σ is unknown (estimated by s)

When to use:

$$T = \frac{\bar{X} - \mu}{s/\sqrt{n}} \sim t_{n-1}$$

We'll use this extensively for confidence intervals and hypothesis tests!

Comparing Normal and t-Distributions

Key differences:

Feature	Normal $N(0,1)$	t-distribution
Shape	Bell-shaped	Bell-shaped
Tails	Thinner	Heavier
Parameters	None (standard)	Degrees of freedom
Use	σ known	σ unknown
Critical value (95%)	1.96	Depends on df

Example critical values for 95% confidence:

- $Z_{0.975} = 1.96$ (always)
- $t_{0.975,df=5} = 2.571$ (small sample)
- $t_{0.975,df=30} = 2.042$ (moderate sample)
- $t_{0.975,df=\infty} = 1.96$ (same as Z)

Other Distributions (Brief Preview)

Chi-Square (χ^2) Distribution:

- Right-skewed, non-negative values
- Used for: goodness-of-fit tests, contingency tables, variance testing
- Parameters: degrees of freedom

F-Distribution:

- Ratio of two chi-square distributions
- Used for: ANOVA, comparing variances
- Parameters: two degrees of freedom (numerator and denominator)

Poisson Distribution:

- Discrete, for rare events
- Used for: disease counts, rare adverse events
- Parameter: rate λ (mean $=$ variance)

We'll encounter these in later weeks!

From Populations to Samples

Fundamental Statistical Problem:

- **Population:** All subjects of interest (usually unknown)
  - Has parameters: μ (mean), σ (SD)
- **Sample:** Subset we actually observe
  - Has statistics: $\bar{x}$ (sample mean), s (sample SD)

Statistical Inference:

- Use sample statistics to learn about population parameters
- Account for sampling variability (uncertainty)
- Quantify our confidence in conclusions

Key Insight: Different samples give different results. How much variation should we expect?

Thought Experiment: OSU Student Heights

Imagine:

- I take a sample of $n = 100$ OSU students, measure heights
- Compute sample mean: $\bar{x}_1 = 66.2$ inches
- **You** take a different sample of $n = 100$ students
- Compute sample mean: $\bar{x}_2 = 65.8$ inches
- **Everyone in class** takes a sample of $n = 100$

Questions:

1. Will all sample means be the same? **No!**
2. Will they be close to each other? **Probably**
3. What's the distribution of all possible $\bar{x}$ values?

Answer: The distribution of $\bar{x}$ is called the *sampling distribution of the mean*

Definition: The probability distribution of all possible sample means from samples of size n

Key Properties (for samples from ANY population):

1. **Center:** $E(\bar{X}) = \mu$ (unbiased estimator)
   - Sample mean centers on population mean
2. **Spread:** $\mathrm{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}$ (standard error)
   - Variability decreases as sample size increases
3. **Shape:** Approximately normal for large n (Central Limit Theorem)
   - Even if population is NOT normal!

Notation: $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ for large n

Standard Error vs Standard Deviation

CRITICAL DISTINCTION:

Standard Deviation (σ):

- Variability of *individual observations*
- Describes spread in the population
- Does NOT decrease with sample size

Standard Error ($\mathbf{SE} = \sigma/\sqrt{n}$):

- Variability of *sample means*
- Describes precision of our estimate
- Decreases as sample size increases
- Smaller SE = more precise estimate

Example: Blood pressure with $\sigma = 20$ mmHg

- Individual SD: $\sigma = 20$ mmHg (always)
- SE for $n = 25$: $20/\sqrt{25} = 4$ mmHg
- SE for $n = 100$: $20/\sqrt{100} = 2$ mmHg

Central Limit Theorem (CLT)

Most Important Theorem in Statistics!

Statement: For a random sample of size n from ANY population with mean μ and finite variance σ^2:

$$\bar{X} \xrightarrow{d} N\left(\mu, \frac{\sigma^2}{n}\right) \text{ as } n \to \infty$$

Or equivalently:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$$

In plain English:

- Sample means are approximately normally distributed
- Works for **ANY** population distribution (skewed, bimodal, etc.)
- Approximation improves as n increases
- Rule of thumb: $n \geq 30$ often sufficient

Why CLT Matters

The CLT enables statistical inference!

Applications:

1. **Confidence Intervals:**
   - We can use normal distribution to quantify uncertainty
   - Even when population is not normal
2. **Hypothesis Testing:**
   - Calculate p-values using normal probabilities
   - Test claims about population means
3. **Sample Size Determination:**
   - Predict precision of estimates before collecting data
   - Plan studies efficiently

Key Insight: We don't need to know the population distribution to make inferences about the mean!

CLT Example: IQ Scores

Population: Los Angeles residents, IQ $\sim N(100, 15^2)$
Actually normal, but CLT still applies!

Sample: $n = 90$ residents, known variance $\sigma^2 = 100$ (so $\sigma = 10$)

Observed: Sample mean $\bar{x} = 105$

Sampling distribution of $\bar{X}$:

- Mean: $E(\bar{X}) = \mu = 100$
- Standard error: $\text{SE} = \frac{10}{\sqrt{90}} = 1.054$
- Distribution: $\bar{X} \sim N(100, 1.054^2)$

Question: How unusual is $\bar{x} = 105$?

- $Z = \frac{105-100}{1.054} = 4.74$ (!!)
- $\Pr(Z > 4.74) < 0.00001$
- Very strong evidence that LA mean IQ > 100

From Point Estimates to Intervals

Problem with point estimates:

- Sample mean $\bar{x} = 105$ is our best guess for μ
- But we know $\bar{x}$ has sampling variability
- How confident are we? What's the range of plausible values?

Solution: Confidence Intervals

- Provide a range of plausible values for μ
- Quantify our uncertainty
- Account for sampling variability

General Form:

$$\text{Estimate} \pm \text{Margin of Error}$$

$$\bar{x} \pm (\text{critical value}) \times \text{SE}$$

95% Confidence Interval (σ known)

When population SD (σ) is known:

$$95\% \text{ CI for } \mu : \quad \bar{x} \pm 1.96 \times \frac{\sigma}{\sqrt{n}}$$

Why 1.96?

- For $Z \sim N(0, 1)$: $\Pr(-1.96 < Z < 1.96) = 0.95$
- Middle 95% of standard normal distribution
- Comes from $\bar{X} \sim N(\mu, \sigma^2/n)$ by CLT

IQ Example: $\bar{x} = 105$, $\sigma = 10$, $n = 90$

$$95\% \text{ CI} = 105 \pm 1.96 \times \frac{10}{\sqrt{90}}$$

$$= 105 \pm 2.07 = (102.93, 107.07)$$

Interpretation: We are 95% confident the true mean LA IQ is between 102.93 and 107.07

95% Confidence Interval (σ unknown)

Reality: We usually don't know σ!

Solution: Use sample SD (s) and t-distribution

$$95\% \text{ CI for } \mu: \quad \bar{x} \pm t_{0.975,n-1} \times \frac{s}{\sqrt{n}}$$

Key differences:

- Use s instead of σ (estimate from data)
- Use t_{n-1} critical value instead of 1.96
- Accounts for extra uncertainty from estimating σ
- **Assumes:** Population is approximately normal

Example critical values:

- $n = 10$: $t_{0.975,9} = 2.262$ (wider than 1.96)
- $n = 30$: $t_{0.975,29} = 2.045$
- $n = 100$: $t_{0.975,99} = 1.984$ (close to 1.96)

Tumor Growth Example

Clinical Trial: 20 patients, 10 treatment, 10 control

Control group tumor growth (mm):
7, 10, 9, 8, 7, 6, 8, 9, 12, 13

Treatment group tumor growth (mm):
4, 6, 10, 8, 5, 3, 10, 8, 8, 10

Control group:

- $\bar{x}_C = 8.9$ mm, $s_C = 2.234$ mm, $n_C = 10$
- $t_{0.975,9} = 2.262$
- 95% CI: $8.9 \pm 2.262 \times \frac{2.234}{\sqrt{10}} = 8.9 \pm 1.598$
- **95% CI: (7.30, 10.50) mm**

Treatment group:

- $\bar{x}_T = 7.2$ mm, $s_T = 2.573$ mm
- 95% CI: **(5.36, 9.04) mm**

Interpreting Confidence Intervals

Correct interpretation:

- "We are 95% confident that the true population mean is between 7.30 and 10.50 mm"
- The interval (7.30, 10.50) either contains μ or it doesn't
- If we repeated this procedure many times, 95% of intervals would contain μ

Common misinterpretations (WRONG):

- $\times$ "There's a 95% probability μ is in (7.30, 10.50)"
  - μ is fixed, not random; it's either in or out
- $\times$ "95% of the data is between 7.30 and 10.50"
  - This is about the population mean, not individual observations
- $\times$ "The sample mean has 95% probability of being in (7.30, 10.50)"
  - We already observed $\bar{x} = 8.9$; no probability involved

Bayesian methods allow probability statements about parameters, but that's beyond this course.

Other Confidence Levels

Not always 95%!

Confidence	α	Z critical	Interpretation
90%	0.10	1.645	Narrower, less confident
95%	0.05	1.96	Standard choice
99%	0.01	2.576	Wider, more confident

Trade-off:

- Higher confidence $\Rightarrow$ wider interval (less precise)
- Lower confidence $\Rightarrow$ narrower interval (more precise)
- Choice depends on context and consequences of error

General formula:

$$\mathrm{CI} = \bar{x} \pm z_{\alpha/2} \times \frac{\sigma}{\sqrt{n}}$$

where $z_{\alpha/2}$ leaves area $\alpha/2$ in each tail

R Example: Binomial Distribution

```
# Vaccine efficacy: n=20, p=0.90
n <- 20
p <- 0.90

# Probability exactly 18 protected
dbinom(18, n, p)  # 0.2852

# Probability at least 18 protected
pbinom(17, n, p, lower.tail=FALSE)  # 0.6769
# or: 1 - pbinom(17, n, p)

# Expected value and variance
expected <- n * p           # 18
variance <- n * p * (1-p)   # 1.8
sd <- sqrt(variance)        # 1.34

# Visualize
x <- 0:n
probs <- dbinom(x, n, p)
```

R Example: Normal Distribution

```
# Cholesterol: mean=200, SD=40
mu <- 200
sigma <- 40

# Probability cholesterol > 240
pnorm(240, mu, sigma, lower.tail=FALSE)  # 0.159

# Probability between 180 and 220
pnorm(220, mu, sigma) - pnorm(180, mu, sigma)  # 0.383

# 90th percentile
qnorm(0.90, mu, sigma)  # 251.3 mg/dL

# Visualize with shaded regions
x <- seq(100, 300, length=200)
y <- dnorm(x, mu, sigma)
plot(x, y, type="l", lwd=2,
     main="Adult Cholesterol Distribution",
     xlab="Cholesterol (mg/dL)", ylab="Density")
```

R Example: Sampling Distribution Simulation

```
# Simulate CLT
set.seed(524)
pop_mean <- 120    # Population mean BP
pop_sd <- 20       # Population SD

# Different sample sizes
sample_sizes <- c(5, 10, 30, 100)

par(mfrow=c(2,2))
for(n in sample_sizes) {
  # Draw 1000 samples, compute means
  sample_means <- replicate(1000, {
    sample <- rnorm(n, pop_mean, pop_sd)
    mean(sample)
  })

  # Plot sampling distribution
  hist(sample_means, breaks=30, prob=TRUE,
       main=paste("n =", n),
       xlab="Sample Mean", col="lightblue")
```

R Example: Confidence Intervals

```
# Tumor growth data (control group)
control <- c(7, 10, 9, 8, 7, 6, 8, 9, 12, 13)

# Method 1: Manual calculation
xbar <- mean(control)                    # 8.9
s <- sd(control)                         # 2.234
n <- length(control)                     # 10
se <- s/sqrt(n)                          # 0.707
t_crit <- qt(0.975, df=n-1)             # 2.262
margin <- t_crit * se                    # 1.598

ci_lower <- xbar - margin                # 7.30
ci_upper <- xbar + margin                # 10.50

cat("95% CI:", round(ci_lower, 2), "to",
    round(ci_upper, 2), "\n")

# Method 2: Using t.test() function
result <- t.test(control, conf.level=0.95)
```

R Example: Comparing Z and t CIs

```
# Sample data
data <- rnorm(15, mean=100, sd=15)
xbar <- mean(data)
s <- sd(data)
n <- length(data)
se <- s/sqrt(n)

# Z-based CI (assumes sigma known = 15)
z_crit <- qnorm(0.975)
z_ci <- xbar + c(-1, 1) * z_crit * 15/sqrt(n)

# t-based CI (sigma unknown, use s)
t_crit <- qt(0.975, df=n-1)
t_ci <- xbar + c(-1, 1) * t_crit * se

# Compare
cat("Z-based CI:", round(z_ci, 2), "\n")
cat("t-based CI:", round(t_ci, 2), "\n")
cat("t CI is", round((t_ci[2]-t_ci[1])/(z_ci[2]-z_ci[1]), 3),
    "times wider\n")
```

Summary

Today we covered:

- **Chapter 7.1:** Binomial distribution
  - Conditions, probability formula, mean and variance
  - R functions: dbinom(), pbinom()
- **Chapter 7.2:** Normal distribution
  - Properties, standardization, empirical rule
  - R functions: dnorm(), pnorm(), qnorm()
- **Chapter 7.4:** Other distributions (t, chi-square, F)
- **Chapter 8:** Sampling distributions
  - Central Limit Theorem
  - Standard error vs standard deviation
  - Confidence intervals (introduction)

Next Week: Chapter 9 - More on Confidence Intervals

Reading: Complete all of Chapter 9

Questions?

Thank you!

Email: john.molitor@oregonstate.edu
Office: 153 Milam